

KEYWAVE

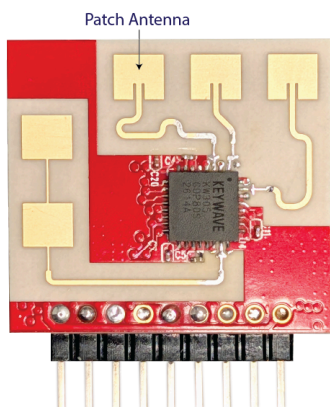
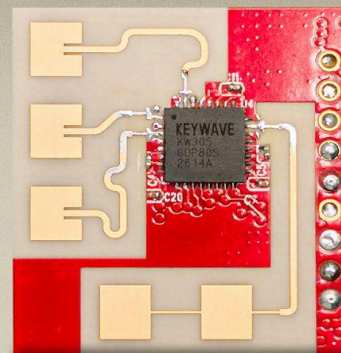
Keywave-KW307

Human Presence Sensor Module

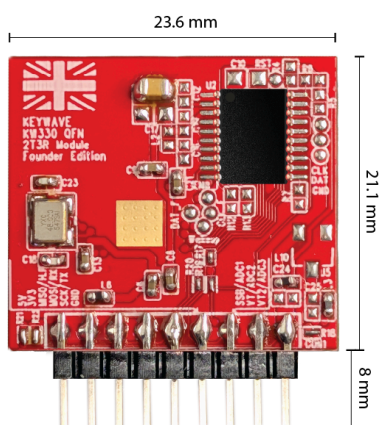
- Revolutionary new Radar technology
- Ultra-low power consumption
- Detects multiple and stationary People
- Reduces false alarms

User Guide and Datasheet

Version 1.03 - 15 th May 2026 Initial release , Keywave



Module Front side
(antenna facing forward)



Module Back Side

Product Overview

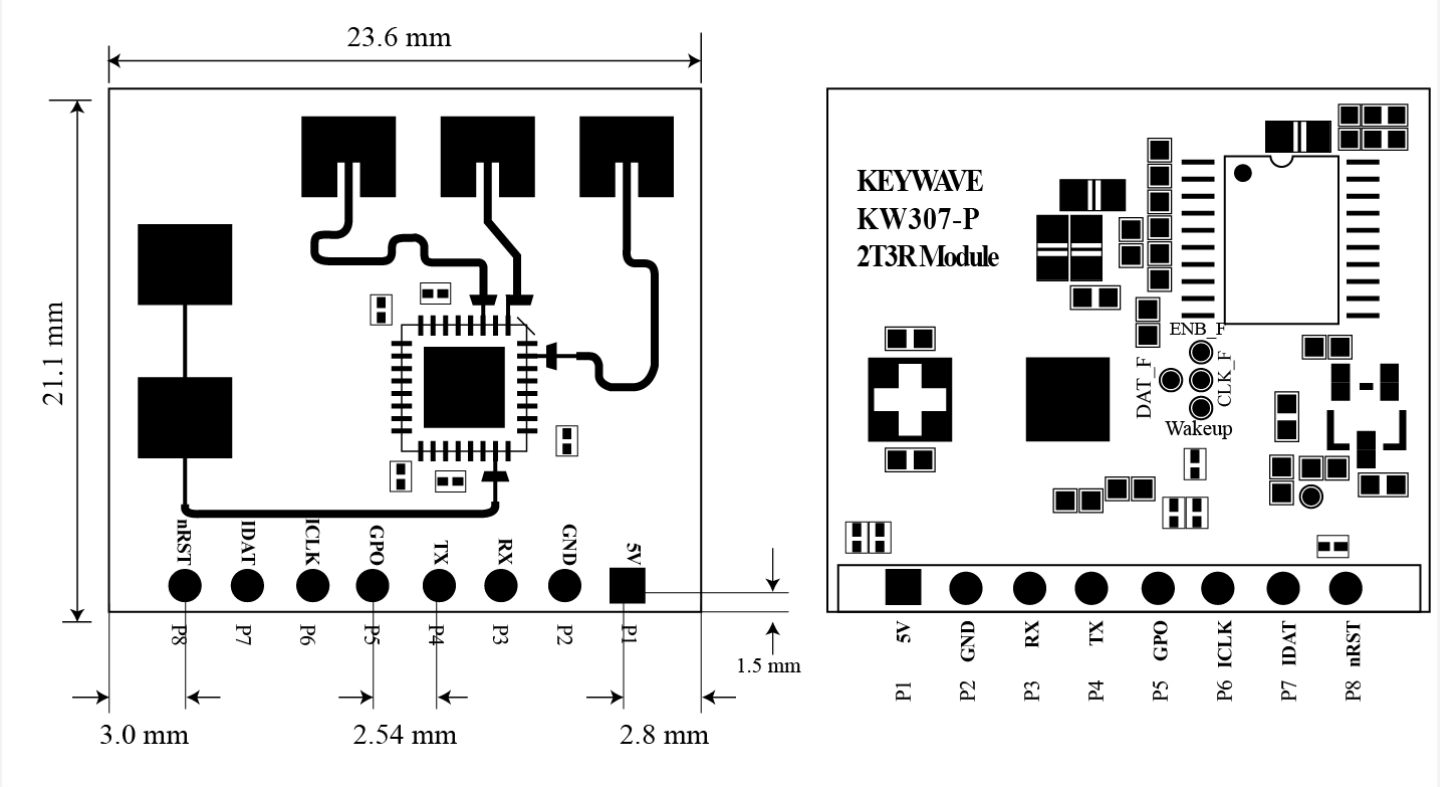
The **KW307** is a compact 24 GHz RF-based **human presence sensor module** designed to detect both moving and stationary people in indoor and outdoor, embedded sensing applications. With integrated antenna, UART target reporting, configurable detection range, adjustable field of view, and support for multi-target tracking of up to three people, the KW307 enables reliable presence-based control for smart buildings, automation, energy-saving systems, and human-machine sensing applications.

Key Features

- 24 GHz RF-based human presence sensing module
- Detects both moving and stationary humans
- Multi-target tracking of up to three people
- Reports target distance and angle information via UART
- Configurable detection range, field of view, gain, sensitivity, and hold time
- Typical 100° field of view, configurable up to 120°
- Distance accuracy up to ± 5 cm and angle accuracy up to $\pm 3^\circ$
- Strong immunity to fan clutter, wind, and vibration
- Supports 3.6V ~ 5V power input
- PC GUI tool for configuration, scenario presets, and real-time detection monitoring

Typical Applications

- Smart lighting and HVAC control, Smart building automation
- Room and desk occupancy detection
- Energy-saving monitoring systems, Smart home and appliance control
- Washroom automation, including urinal and hand dryer sensing
- Human-machine interface and embedded presence detection
- Evaluation and prototyping for RF sensing products



Pin	Symbol	Type	Description
P1	5V	Power	5V power supply
P2	GND	Ground	Ground
P3	RX	Input	UART Rx signal
P4	TX	Input	UART Tx signal
P5	GPO	Output	General-purpose output
P6	ICLK	Input	Programming clock signal
P7	IDAT	Input	Programming data signal
P8	nRST	Input	System reset signal (active low)

- 5V Power Mode: Supply voltage is regulated by an on-chip LDO to power the internal core.

Electrical Specification

Absolute Maximum Ratings

Characteristic	Symbol	Min.	Typ.	Max.	Units
5V supply voltage	V_{DD5V}	-0.3	-	6.0	V
Digital I/O voltage	V_{DIO}	-0.3	-	3.6	V
Ambient Temperature	T_A	-25	-	85	°C

DC Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Supply voltage via 5V pin	V_{DD5V}	3.6	5.0	6.0	V
Operation Current @ 20Hz update rate	I_{DC}	-	3.0		mA

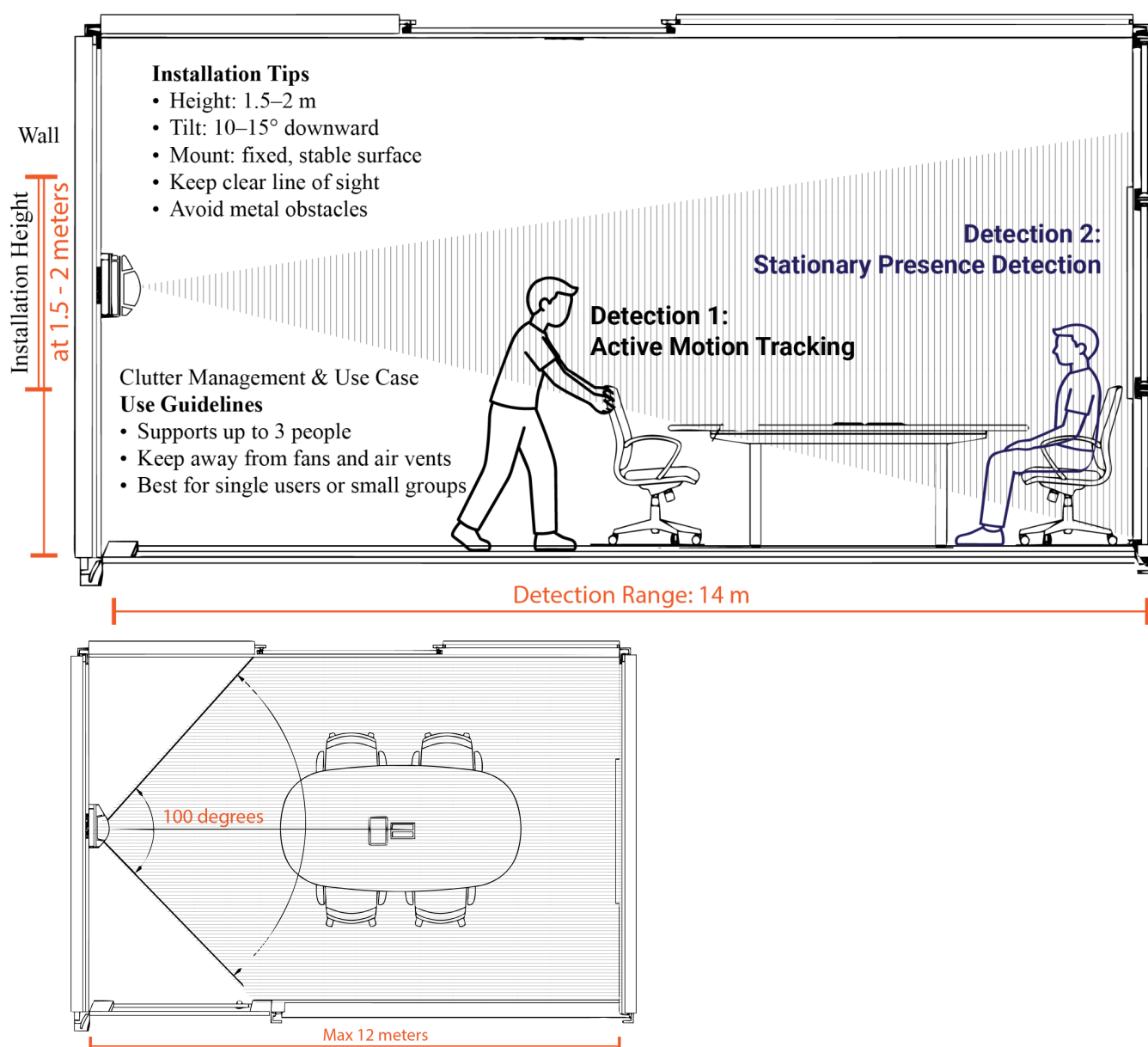
RF Characteristics

Characteristic	Symbol	Min.	Typ.	Max.	Units
Operation frequency	F_{RF}	24.05	-	24.25	GHz
RF output power	P_{RF}	-	-2	-	dBm
Detection range	Range	-	-	14	m
Field of View	FOV	-	100	-	degrees

*Horizontal alignment at 1m height.

Mounting Suggestion

- Optimal Installation Height:** For best results, mount the sensor at a height of 1.5m to 2m above the floor.
- Avoid Obstructions:** Ensure a clear line of sight. Avoid placing the sensor behind metallic objects or thick glass, as these may block or attenuate the radar signal.
- Directional Alignment:** If the target area is close to the mounting point tTilt the sensor slightly downward (approx. 10° – 15°) to maximize FOV coverage.
- Stability:** Mount the sensor on a rigid surface. Although the KW307 has superior immunity to vibration, a stable mount ensures the highest possible accuracy for stationary detection.
- Clutter Management:** Position the sensor away from large moving mechanical parts (such as large industrial fans) to maintain the best tracking performance for multiple people.

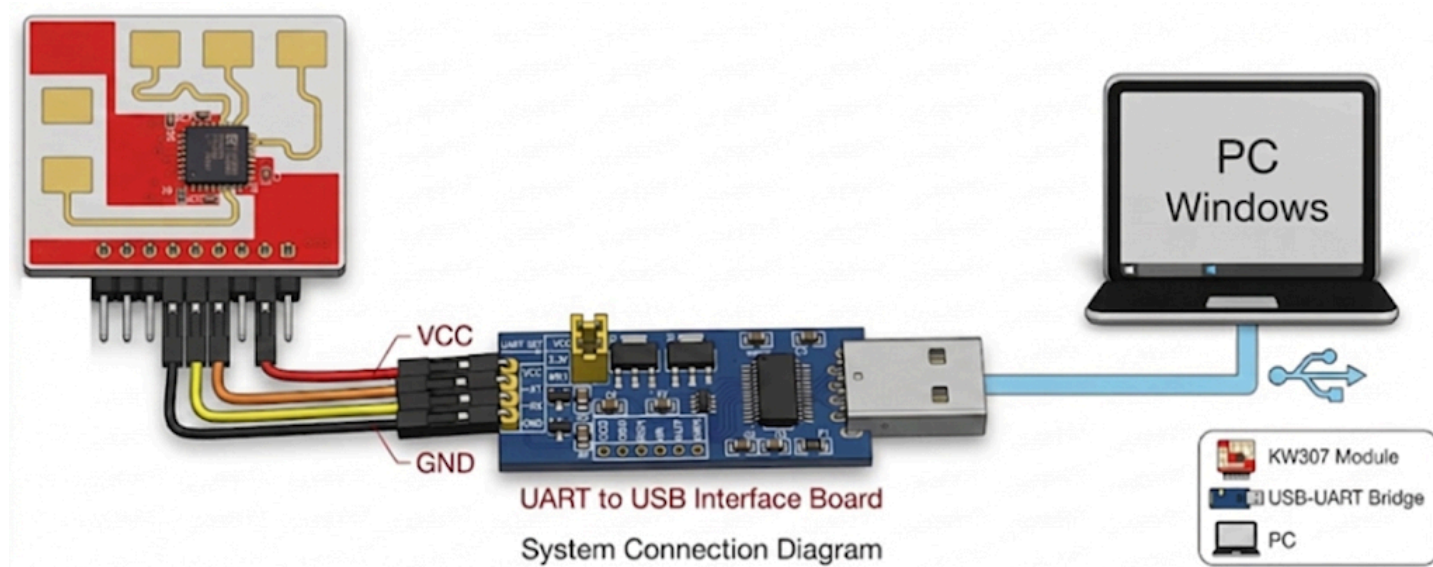


Configuration and monitoring tool

The KW307 configuration and monitoring tool is a GUI application designed to help you get up and running as fast as possible.

The tool allows you to:

- Configure sensor detection parameters, making it easy to tune the KW307 to your application
- Choose pre-set tuning parameters for common applications.
- Monitor real-time detection status, on an intuitive graphical display.



- The configuration tool uses a USB to UART adapter to communicate from a Windows PC to the KW307.
- Note the Pin spacing for the KW307 is 2.54mm .

CAUTION :

- When connecting the adaptor, or bridge board, ensure you use a data-capable USB cable, not a charge only cable.
- Take care when connecting the cable as connection errors may permanently damage the hardware module.
- The configuration tool software can be found at : <https://keywavetech.co.uk/products>

GUI Configuration Overview

Step1 : Select "COM Port"

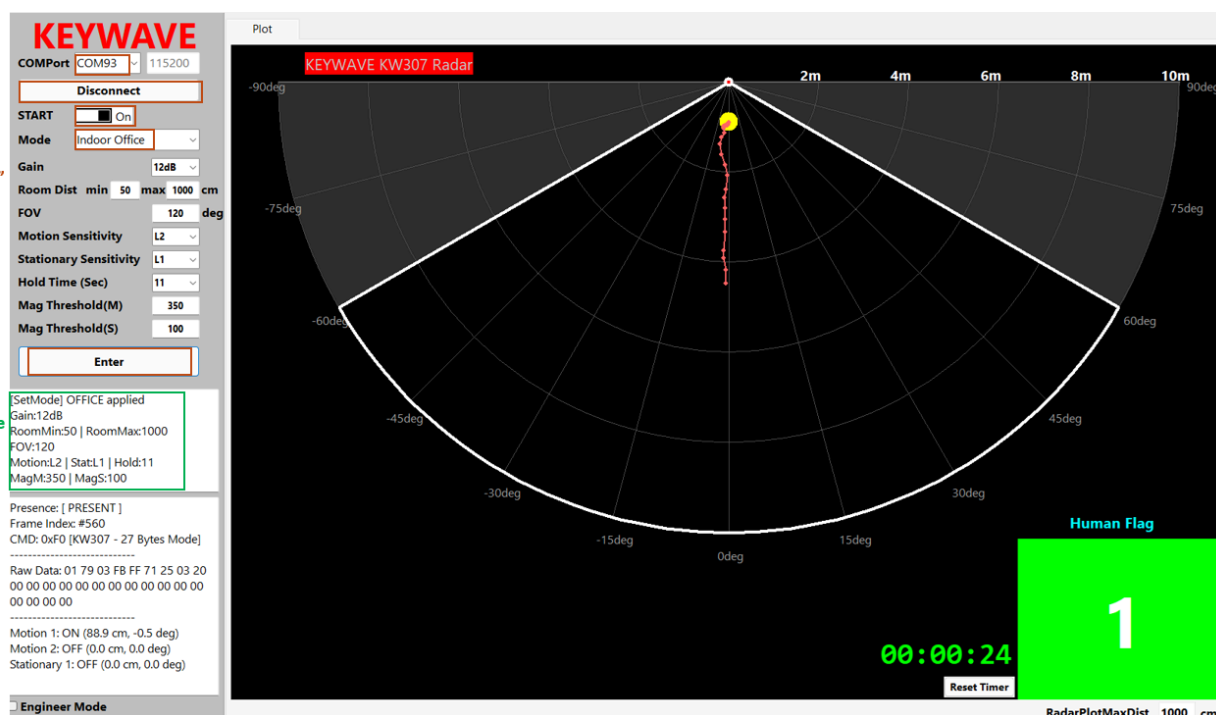
Step2 : Click "Connect"
(to connect com port)

Step3 : Click "START On"
(Sensing start)

Step4 : Select "Scenarios"
(Setup the setting)

Step5 : Click "Enter"

Module
Active



- The pre-loaded application scenarios comes with pre-defined optimal parameters, while still allowing users to fine-tune settings for specific applications..

Function	Range	Description
Application Mode	5 scenarios	Outdoor, Indoor_office, Desk, Urinal, Hand_dryer
Gain	-3db ~ +21db	Adjust radar signal strength.
Room Dist (min/max)	10 ~ 1400	Set the range size for radar detection. (cm)
FOV	0 ~ 120	Set the radar field of view. (degree)
Motion Sensitivity	L0 ~ L9	The ability to detect a moving human. L0: Most sensitive (i.e., Wave hands or shake heads) L9: Least sensitive (i.e., Walk one step)
Stationary Sensitivity	L0 ~ L9	The ability to detect subtle movement of a stationary human. L0: Most sensitive (i.e., Breath) L9: Least sensitive (i.e., Wave hands or shake heads)
Hold Time	1 ~ 60	Set the hold time of Human Flag as human leaves detection area (Sec).
Mag Threshold (M)	0 ~ 1000	Signal strength above the mag threshold will be identified as valid targets for a moving human. (Environment-dependent, default : 350)
Mag Threshold (S)	0 ~ 1000	Signal strength above the mag threshold will be identified as valid targets for a stationary human. (Environment-dependent, default : 100)

Ordering Information

Contact: Q@keywavetech.co.uk

Item	Description
Product	KW307 Human Presence Sensor Module
Part number	KW307
Module variant	KW307-P
Product name	KW307 Human Presence Sensor Module
Sensor type	24 GHz RF-based human presence sensor module
Interface	UART, GPO, programming interface pins
Supply options	5 V input or 3.3 V input; connect only one supply at a time
Package / module type	PCB module with integrated antenna
Module dimensions	Approximately 23.6 mm × 21.1 mm
Pin header	2.00 mm pitch on KW307 module
Firmware version	TBD / To be confirmed
Hardware revision	TBD / To be confirmed
Datasheet version	v1.0
Evaluation support	PC GUI configuration and monitoring tool

For the latest updates, visit our website: www.keywavetech.co.uk

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